CNN Design of Micro-Circulation Bus Routes in Scenic Areas

Soldier[a]

[a]*School of Mathematics and Statistics, Henan Normal University, Xinxiang, 453007, P.R. China*

Abstract

This paper develops a multi-period optimization model for the design of micro-circulation bus routes in scenic areas. The model jointly determines node construction, period-specific node activation, route opening, and route topology. The objective integrates passenger service cost, operator cost, an off-peak service incentive, and a temporal demand-balance penalty. The resulting formulation is a mixed-integer quadratic optimization model.

Keywords: Micro-circulation bus, Scenic area transit, Route design, Multi-period optimization, Mixed-integer quadratic programming

---

1. problem description

We address the multi-period route design problem for buses shuttle services in touristic sites areas. Let $G(J, A)$ be a graph wherein $J = \{1, \ldots, n\}$ be the set of all nodes including candidate scenic spots and parking lots and A is the set of all possible directed connections between every pair of node $(v_1, v_2) \in J \times J$. We define $D = \{1, \ldots, d\} \subseteq J$ to denote the set of scenic spots only. Let $T = \{0, 1, \ldots, T_{\max}\}$ be the set of discrete time periods. For each node $j \in J$ and period $t \in T$, $q_{j,t} \geq 0$ represents the passenger demand at node j in period t. The travel time between any two nodes $(j, l) \in A$ is given by the known constant $t_{jl} \geq 0$. Each route $k \in K$, where $K = \{1, \ldots, m\}$ is an ordered subset of nodes in J, has a capacity $C_k > 0$, which is the maximum number of passengers that can be carried in a single run of the route. Each shuttle bus has a capacity C^b and the number of busses deployed on a route is $\lceil C_k/C^b \rceil$. Each scenic point has a capacity C^s. The set of routes K is an exogenous part of the problem and is a parameter. We would like to jointly determine, for each period $t \in T$, the routes to be offered to the visitors such that the capacity of every scenic point is respected given the number of routes visiting that point at t and the number of visitors on every route at t. The objective is to design routes in such a way to minimize the total system operating cost (number of busses) while having a fair capacity utilisation (min(volume − capacity)^2, variance) across all nodes at every period along the planning horizon.

2. Mathematical Formulation

2.1. Sets

- $J = \{0, 1, \ldots, 14\}$: Set of all nodes.
- $P = \{0, 1, 2\}$: Set of parking lots.
- $K = \{0, 1, 2\}$: Set of routes.
- $T = \{1, 2\}$: Set of time periods.
- $A = \{(i, j) \mid i, j \in J, i \neq j\}$: Set of feasible arcs.

2.2. Parameters

- D_{ij}: Travel distance from node i to node j.
- demand$[t-1, p]$: Demand associated with parking lot p in period t.
- C_j: Capacity of node j (defined for $j \notin P$).

2.3. Decision Variables

- $y_{ijkt} \in \{0, 1\}$: Binary variable indicating whether route k uses arc (i, j) in period t.
- $h_{ikt} \in \{0, 1\}$: Binary variable indicating whether route k visits node i in period t.
- $f_{ijt} \geq 0$: Continuous variable representing the total flow on arc (i, j) in period t.

2.4. Objective Function

$$\min \sum_{i \in J} \sum_{j \in J} \sum_{k \in K} \sum_{t \in T} D_{ij} \cdot y_{ijkt} \tag{1}$$

2.5. Constraints

2.5.1. Flow Balance Constraints (on binary variables)

$$\sum_{\substack{j \in J \\ j \neq i}} y_{jikt} = \sum_{\substack{j \in J \\ j \neq i}} y_{ijkt}, \quad \forall i \in J,\, k \in K,\, t \in T \tag{2}$$

2.5.2. Node Degree Constraints for Non-Parking Nodes

$$\sum_{k \in K} \sum_{\substack{j \in J \\ j \neq i}} y_{ijkt} \geq 1, \quad \forall i \in J \setminus P,\, t \in T \tag{3}$$

$$\sum_{k \in K} \sum_{\substack{j \in J \\ j \neq i}} y_{jikt} \geq 1, \quad \forall i \in J \setminus P,\, t \in T \tag{4}$$

2.5.3. Parking Lot Usage Limit

$$\sum_{k \in K} h_{pkt} \leq 2, \quad \forall p \in P,\, t \in T \tag{5}$$

2.5.4. Linking Constraints between Arc Usage and Node Usage

$$y_{ijkt} \leq h_{ikt}, \quad \forall (i,j) \in A,\, k \in K,\, t \in T \tag{6}$$

2.5.5. Anti-Parallel Arc Constraints

$$y_{ijkt} + y_{jikt} \leq 1, \quad \forall (i,j) \in A,\, k \in K,\, t \in T \tag{7}$$

2.5.6. Route Length Constraints

$$\sum_{(i,j) \in A} y_{ijkt} \leq 6, \quad \forall k \in K,\, t \in T \tag{8}$$

2.5.7. Departure from Parking Lots

$$\sum_{p \in P} \sum_{\substack{j \in J \\ j \neq p}} y_{pjkt} = 1, \quad \forall k \in K,\, t \in T \tag{9}$$

2.5.8. Return to Parking Lots

$$\sum_{p \in P} \sum_{\substack{j \in J \\ j \neq p}} y_{jpkt} = 1, \quad \forall k \in K,\, t \in T \tag{10}$$

2.5.9. Each Parking Lot Must Be Used Exactly Once (Return)

$$\sum_{k \in K} \sum_{\substack{j \in J \\ j \neq p}} y_{jpkt} = 1, \quad \forall p \in P,\, t \in T \tag{11}$$

2.5.10. Flow Lower Bound Constraints from Parking Lots

$$\left(\sum_{k \in K} y_{pjkt} \right) \cdot \text{demand}[t-1, p] \leq f_{pjt}, \quad \forall p \in P,\, j \in J \setminus P,\, t \in T \tag{12}$$

2.5.11. Flow Lower Bound Constraints to Parking Lots

$$\left(\sum_{k \in K} y_{jpkt} \right) \cdot \text{demand}[t-1, p] \leq f_{jpt}, \quad \forall p \in P,\, j \in J \setminus P,\, t \in T \tag{13}$$

2.5.12. Flow Conservation Constraints (Non-Parking Nodes)

$$\sum_{\substack{i \in J \\ i \neq j}} f_{jit} = \sum_{\substack{i \in J \\ i \neq j}} f_{ijt}, \quad \forall j \in J \setminus P,\, t \in T \tag{14}$$

2.5.13. Big-M Constraints Linking Flow and Route Selection

$$f_{ijt} \leq M \sum_{k \in K} y_{ijkt}, \quad \forall (i, j) \in A,\, t \in T \tag{15}$$

where $M = 100$ is a sufficiently large constant.